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MATHEMATICS 2210
Quiz 6, Fall semester 2012

Name: KEY

1. (6 pts) Find maxima and minima of $f(x,y) = x^2 - 2y^2$ on $S = \{(x,y) | x^2 + y^2 \le 1\}$.

$\nabla f = (2x, -4y)$ $\nabla f = 0 \Rightarrow x = 0 \quad y = 0$ $f(0,0) = 0 \leftarrow$ saddle

$\partial S = \{x^2+y^2=1\}$ $x = \cos\theta$, $y = \sin\theta$ $f(\theta) = \cos^2\theta - 2\sin^2\theta = 1 - 3\sin^2\theta$

$f'(\theta) = -3\sin\theta\cos\theta$

$f' = 0$ if $\theta = 0, \pi/2, \pi, 3/2\,\pi$ or

$(x,y) = (1,0), (0,1), (-1,0) \ (0,-1)$

$f(1,0) = f(-1,0) = 1$

$f(0,-1) = f(0,1) = -2$

Max $f(1,0) = f(-1,0) = 1$

MIN $f(0,-1) = f(0,1) = -2$

2. (6 pts) Find the critical points of $f(x,y) = x^3 + y^3 - 6xy$ and use the second derivative test to determine if they are local maxima, local minima, saddles or undetermined.

$\nabla f = (3x^2 - 6y, 3y^2 - 6x)$ $\begin{cases} 3x^2 - 6y = 0 \\ 3y^2 - 6x = 0 \end{cases}$ $\begin{cases} x^2 = 2y \\ y^2 = 2x \end{cases}$ $x^4 = 4y^2 = 8x$

$x(x^3 - 2) = 0 \to x = 0 \to y = 0$

$\to x = 2 \to y = 2$

$D = f_{xx}f_{yy} - f_{xy}^2 = 6x \cdot 6y - (-6)^2 = 36(xy - 1)$

$D(0,0) = -36$ saddle

$D(2,2) = 108 > 0$, $f_{xx}(2,2) = 12 > 0 \Rightarrow$ Local min.